Jacobs Baymax Caregiver Support Robot

Mission and first year scope | R0.1 | 23 October 2026
Accountable owner Faisal Farouq | Provisional planning baseline

Mission. Assist caregivers with companionship, engagement, fictional record organization and communication while caregivers retain responsibility, authority and clinical judgment. The robot identifies itself as a research support system and claims no professional-care or emergency-response authority.

Physical research goals. Target height 1.829 m and operating mass no more than 55 kg, including battery, rigid panels and attached modules. Above 55 kg requires rework; a projection above 65 kg requires major design review. Use 10 active lower-body joints. Complete three consecutive tethered steps in each of two supervised trials, targeting 0.10-0.15 m/s, inside a cleared controlled area with rated robot fall protection.

Caregiver software. Develop the laptop demonstration first. Use fictional profiles and simulated readings only. Demonstrate games and conversation, consented preferences, five accessibility/age presets, observation trends, medication-response logs, audit history, summaries and approved-recipient telepresence. Every reading shows value, units, source and time. Every non-emergency report needs exact-content preview and explicit confirmation. Receipt is distinct from human review.

Separate handling goal. Move a 2 kg inert object through an approved controlled trajectory for 10 supervised repetitions while stationary and mechanically secured. Log position, current, temperature and stop events. Never walk with the test object. Never lift, catch, transfer, restrain or support a person.

Appearance. Follow the written visual baseline: friendly rounded proportions, rigid removable white plastic panels, narrow visible seams, protected charcoal joint gaps, subtle teal, two connected circular eyes and broad flat rounded-rectangle feet. Cooling, real motion, service access, sensing, fall protection and stop access constrain styling. No inflatable, combat, superhero or military exterior.

Boundaries. No real patient information, live EHR, diagnosis, clinical scan interpretation, treatment or dose advice, medication selection/dispensing/administration, unsupervised high-energy motion or unapproved public operation. The safety controller remains independent from conversational AI. A 5 kg payload, clinical subsystem, untethered gait and cargo walking require later reviews.

Feasibility and release. Retain the USD 5000-10000 funding range and USD 10000 ceiling. The ceiling allocation is unquoted; an illustrative commercial-actuation route totals USD 15899. The 55 kg total is an allocation rather than a measured prediction. No system tests, simulation runs, fabrication, professional approvals or purchases are claimed. Major procurement is BLOCKED until architecture, sizing, simulation, quotes, independent review, named fabrication and inspection gates are satisfied.

Acceptance and next action. Full first-year completion requires reviewed evidence for software, both gait trials, supported handling and honest limitations. Resolve D-01 funding feasibility, D-02 facility and qualified people, D-03 joint topology, D-04 handling arrangement and D-05 fictional workflow and approved recipient. Document coverage: PASS. Owner acceptance: pending. No physical work is authorized by this page. Budget change approved: USD 0.